

Why this is not a party trick

Roughly half of contacts-v1 pretraining is sequence modelling. From exp53: 4,213,203 documents, ~4.67B train tokens, mean 1,131 tokens/doc, mean ~200 contacts/doc. Contacts cost 3 tokens each and the frame is 4, so the sequence section is ~527 tokens — about 47% of all training tokens, half of which are the <AA> targets. exp199's 152B-token run therefore spent on the order of 35B tokens predicting amino acids from other amino acids.

There is no loss masking: marifold_models trains a plain LM loss over packed documents, so the sequence section is inside it.

The default model was trained as an any-order model on purpose. exp199's winner is its "-aug" arm, which the exp166 sweep defines as giving every packed example a fresh re-permutation of its <pN> <AA> statements, ramped from probability 0 to 1 over training. Under a uniform shuffle any statement can be last, so asking for $P(\text{residue } i \mid \text{all the others})$ asks something the model saw directly.

One pass per ordering, not one per residue

A document is an ordinary causal-LM sequence, so a single teacher-forced pass yields the conditional at every residue at once: the slot holding a <pN> token already carries $P(\text{amino acid at N} \mid \text{every statement before it})$. An L-residue protein costs one forward pass per ordering rather than L.

What each residue is conditioned on varies with where its statement landed in the shuffle, so the readout also returns a per-slot context size. That turns what would be a nuisance into two knobs no masked LM has: how many orderings to ensemble, and how much context to require.

The whole 212-assay benchmark took 29.1M tokens and 54.5 minutes on one A5000, at 8,912 tokens/sec. The cost forecast in the issue said 29.1M tokens.

Preflight: three ways this could have lied

The aggregation is exact. ProteinGym's headline is not a mean over assays; it is a mean within UniProt id, then within function category, then over the five categories. Our implementation reproduces every published leaderboard number from their own per-assay file to within 0.0005.

Mutation indexing is right. Across 212 assays and 2,438,361 variants, the stated wild-type letter matches target_seq at the stated position every time.

The rope config survives the load. The bucket copy of exp199 carries both transformers-4 and transformers-5 rope, so the silent theta-10000 fallback, worth 0.76 nats/token, does not fire.

Phase 0: the conditional is real

Top-1 accuracy at context ≥ 0.8 is 0.345, against a 0.132 composition floor and 0.088 for a scrambled-sequence control. Perplexity 8.15 versus 18.72.

The scrambled arm is the load-bearing control. It stays flat as context grows (0.094 to 0.088) while the model arm nearly triples (0.103 to 0.345). Composition is identical in both arms by construction, so everything the model gains from seeing more of the protein is real sequence structure.

The conditional is also soft rather than memorised: $P(\text{wt})$ is only 0.231 at high context, with entropy 2.15 of a possible 3.00 nats. So the ranking of the other 19 amino acids, which is the only thing a variant-effect score reads, sits in a broad distribution. An early ubiquitin smoke test had returned top-1 0.98, which would have implied a memorised spike; across 14 diverse proteins that is an outlier.

Phase 1: both knobs work, and double the score

K=1 with every slot scores 0.147. K=200 with context ≥ 0.9 scores 0.2964.

The ordering ensemble saturates fast: most of its gain is in by K=16 (0.147 to 0.225), and K=64 to K=200 buys 0.0011. The context threshold keeps paying to the end. The pre-registered primary rule is also the grid's best cell, so nothing here rests on test-set selection.

The readout approximation is not what costs us the gap. At context ≥ 0.9 a residue is never conditioned on exactly all-but-one, but the context increments are diminishing (+0.038, +0.018, +0.009), so exact masked marginals is worth 0.01 or less against a 0.119 gap. The gap is model quality, not estimator error.

The structure-tilt prediction fails

The issue predicted a structure-trained model would tilt toward Stability. It does. So does an equally weak sequence-only model, which is the control that makes the prediction falsifiable.

MarinFold scores Stability at 1.36x its own average and OrganismalFitness at 0.71x. ESM2-35M, a sequence-only model at a matched overall level, scores 1.36x and 0.71x. ESM-IF1, a genuine structure model, scores 1.47x.

MarinFold's profile is indistinguishable from ESM2-35M's and unlike ESM-IF1's. Whatever this model has learned presents as a small sequence model, not a structure model wearing a sequence readout. The Stability tilt is a property of weak variant-effect predictors generally, not evidence of structural knowledge.

Where it wins and where it is worst

MarinFold beats ESM-2 650M on 34 of 212 assays, 16%, median delta -0.106.

Best stratum is Stability, 29% win rate, 0.403 against 0.523. Worst are OrganismalFitness at 7% and viruses at 7%.

Viruses are the standout weakness: 0.111 against ESM-2's 0.272. That is worth flagging because the prior ran the other way — exp199 trains on ESM-Atlas metagenomic data, so one might have expected better generalisation to under-represented sequence space. It does not materialise.

Does it add anything on top of ESM-2? No.

Per-assay correlation with ESM-2 is $r = 0.696$, loose enough that the two might have failed on different assays — which would make a weaker model still worth having. Joining ProteinGym's per-variant archive to our conditionals and combining by within-assay rank settles it.

Weight on Mariner's rank, against average Spearman: 0.0 gives 0.4152, 0.1 gives 0.4170, 0.2 gives 0.4162, 0.3 gives 0.4120, 0.5 gives 0.3903, 1.0 gives 0.2964.

The pre-registered equal-weight ensemble scores 0.3903, a lift of -0.0249. Mixing Mariner in at 50% makes ESM-2 measurably worse (bootstrap -0.0256, 95% CI -0.0328 to -0.0191). The best cell is $w=0.1$ at +0.0016, reproducibly positive and negligible: smaller than the 0.0023 noise floor #204 measured on one unchanged checkpoint, and selected on the benchmark besides.

The decorrelation does not translate into complementary signal. Mariner is not a weak-but-useful ensemble member; its errors overlap ESM-2's enough that adding it dilutes. This also cross-validates the pipeline: the curve's endpoints come from per-variant archive data through a different code path and reproduce both headline numbers exactly, 0.2964 at $w=1$ and 0.4152 at $w=0$.

Conclusion and what is next

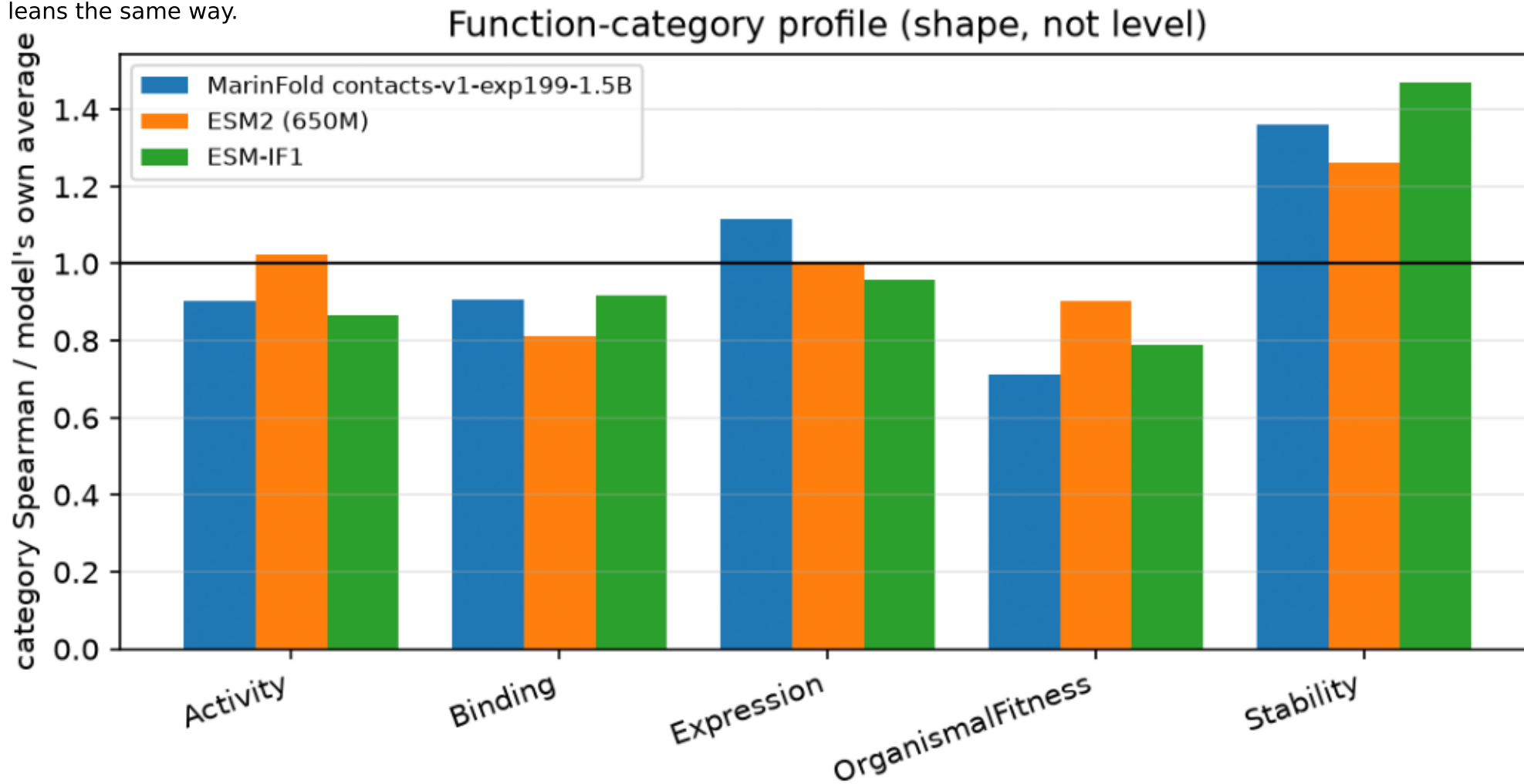
contacts-v1 is a real but weak bidirectional protein language model. The conditional is genuinely contextual and converts to 0.2964 zero-shot Spearman, well clear of the 0.188 broken-readout floor — but that is ESM2-35M territory.

The more interesting finding is the negative one, and it has two independent legs. On the function-category profile MarFold is indistinguishable from a small sequence-only model. And it adds nothing in an ensemble with ESM-2, despite the loose correlation. Training on a structure objective bought sequence understanding roughly in proportion to the sequence tokens seen, and nothing extra this benchmark can detect.

Two things stay open and are both cheap now the harness exists. Exact joint scoring for multi-mutants: 1.77M of 2.44M variants are multi-mutant and every single-sequence baseline scores them additively. And the p03-base versus p03-aug ablation, exp199's matched pair that ties on contact R-precision, where ProteinGym measures exactly what the order augmentation was for.

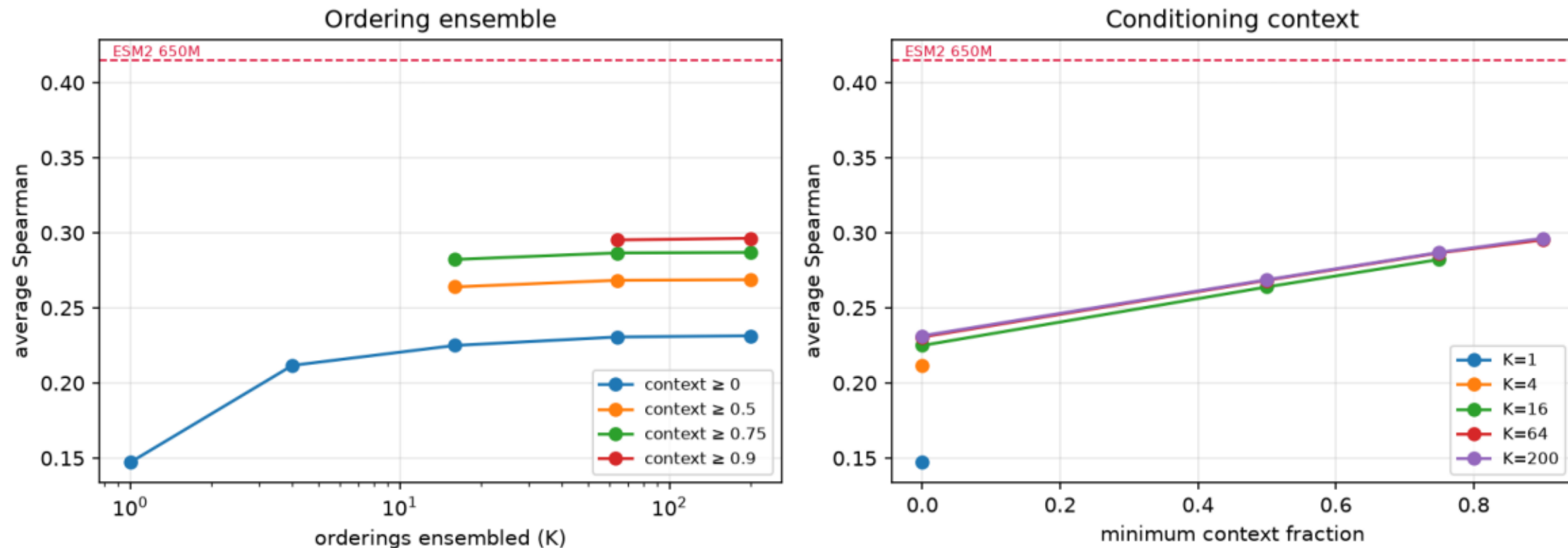
category_profile.png

Each model's per-category Spearman divided by its own overall average, so the bars compare *shape*. A structure-trained model is expected to lean on Stability (ESM-IF1 does); the question is whether MarFold, trained on a structure objective but reading out sequence, leans the same way.



sweeps.png

Average Spearman on the 212 in-range ProteinGym substitution assays vs the two knobs an any-order model has and a masked LM does not: how many document orderings are ensembled (left) and how much of the rest of the protein each conditional saw (right). Dashed line is ESM-2 650M re-aggregated on the same 212 assays.



vs_esm2.png

Per-assay Spearman, MarFold vs ESM-2 650M, on the 212 shared assays. Points below the diagonal are assays ESM-2 wins. A low correlation would mean the two models fail on different assays — which is what would make a structure-objective model useful in an ensemble even at a lower overall level.

